

Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees. Each tree is trained on a randomly sampled subset of the training data and features, and the final class is determined by majority voting.

Random Forest

In this section, a Random Forest classifier is constructed, and its identification performance is evaluated on the above two datasets.

Click to run.

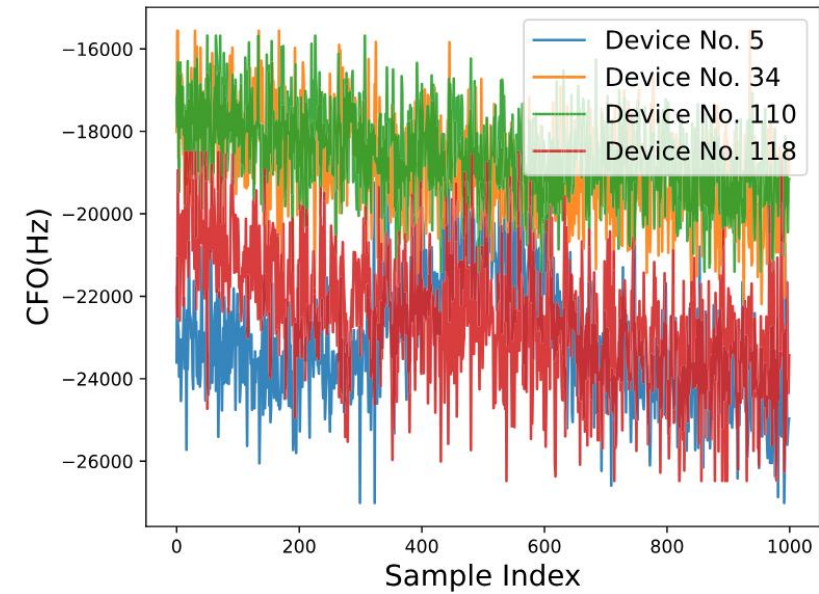
```
import os
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
from mpl_toolkits.axes_grid1.inset_locator import inset_axes
import seaborn as sns

plt.rcParams.update({'font.size': 12})
sns.set_context("notebook", font_scale=1.5)
```

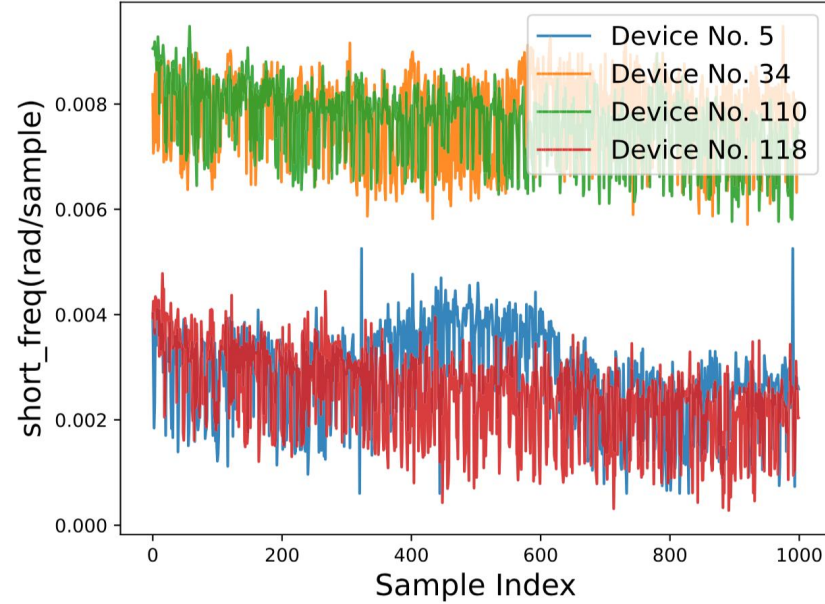
Evaluation results for random forest model.

Dataset & Setting	Accuracy (ACC)
5-Fold Average without KF	82.0
5-Fold Average with KF	88.6

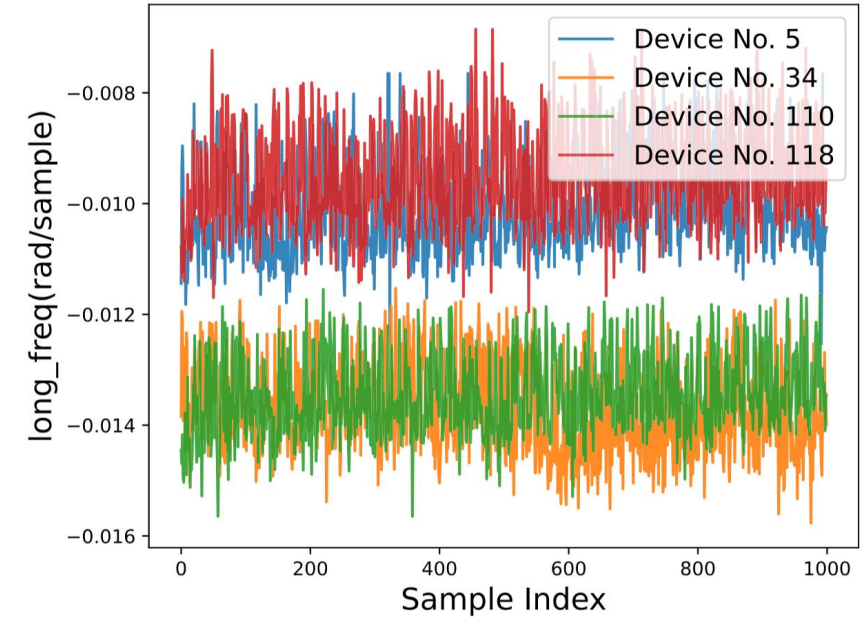
Impact of sessions on representative RF features



(a) CFO



(b) Coarse CFO of STS



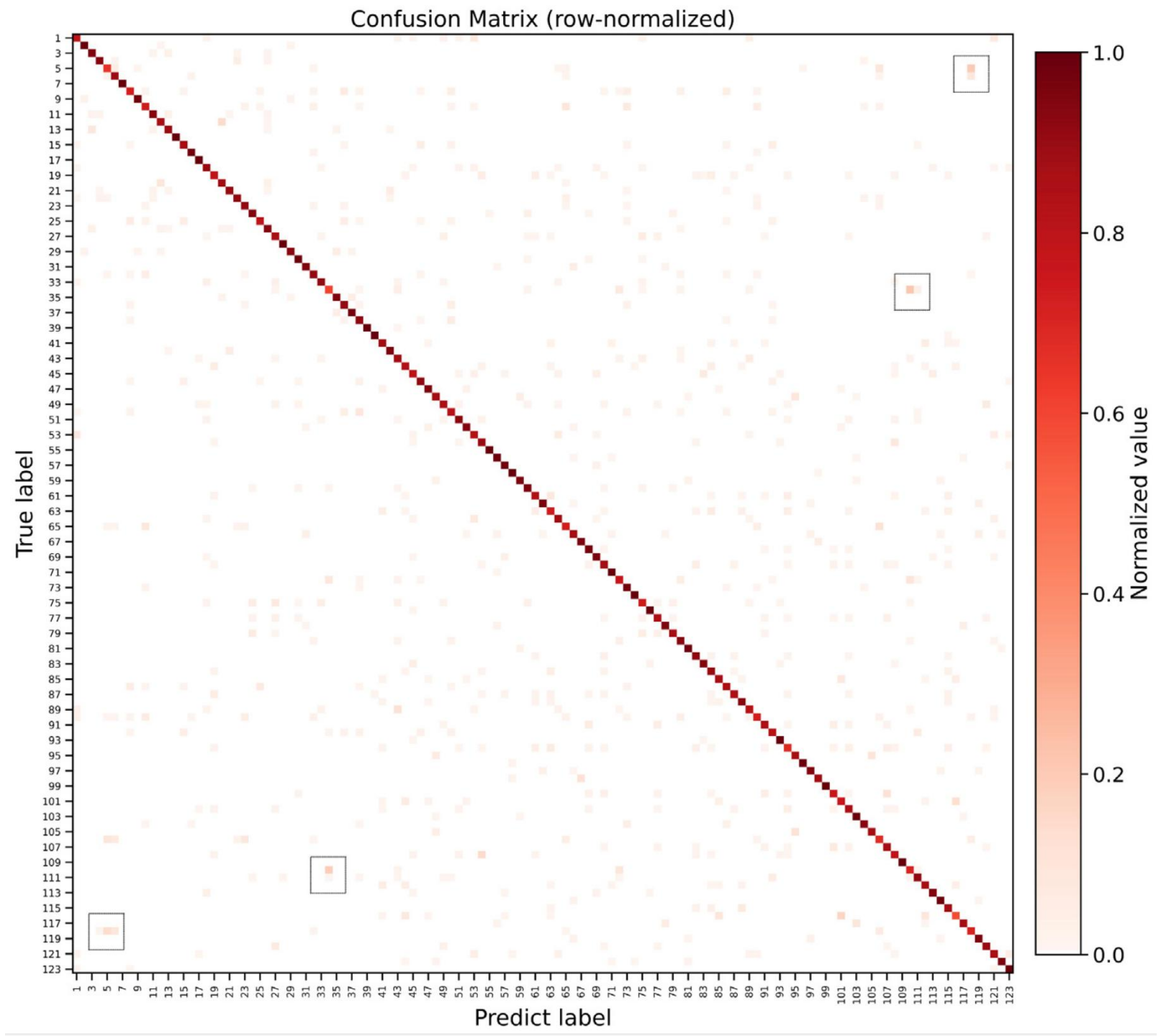
(c) Fine CFO of LTS

Feature importance

Feature importance for random forest model.

Notation	Feature	Importance
f_1	CFO	0.2199
f_2	Coarse CFO of STS (short_freq)	0.1760
f_3	Fine CFO of LTS (long_freq)	0.1442
f_4	Mean of phase error vector of L1 (phase_error_mean_1)	0.0562
f_5	Mean of phase error vector of L2 (phase_error_mean_2)	0.0544
f_6	Variance of phase error vector of L2 (phase_error_var_2)	0.0418
f_7	Variance of phase error vector of L1 (phase_error_var_1)	0.0417
f_8	I/Q gain imbalance of L1 (iqi_1)	0.0382
f_9	Mean of magnitude error vector of L2 (mag_error_mean_2)	0.0381
f_{10}	Mean of magnitude error vector of L1 (mag_error_mean_1)	0.0375
f_{11}	I/Q gain imbalance of L2 (iqi_2)	0.0375
f_{12}	Variance of magnitude error vector of L1 (mag_error_var_1)	0.0305
f_{13}	Variance of magnitude error vector of L2 (mag_error_var_2)	0.0305
f_{14}	Fractal dimension of L1 (frac_dimension_1)	0.0268
f_{15}	Fractal dimension of L2 (frac_dimension_2)	0.0265

Confusion Matrix



Identification accuracy versus feature groups

